

Python 3.13.13 (tags/v3.13.13:01104ce, Apr 7 2026, 19:25:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

```
===== RESTART: C:/Users/DELL/AppData/Local/Programs/Python/Python313/qn1.py =====  
Water Jug Solution Path (jug1, jug2):
```

```
Step 0: (0, 0)  
Step 1: (0, 3)  
Step 2: (3, 0)  
Step 3: (3, 3)  
Step 4: (4, 2)
```

```
===== RESTART: C:/Users/DELL/AppData/Local/Programs/Python/Python313/qn1.py =====
```

```
Action: navigate_around | Position: (0, 0) | Battery: 96 | Samples: 0  
Action: navigate_around | Position: (0, 0) | Battery: 92 | Samples: 0  
Action: collect_sample | Position: (0, 0) | Battery: 87 | Samples: 1  
Action: collect_sample | Position: (0, 0) | Battery: 82 | Samples: 2  
Action: collect_sample | Position: (0, 0) | Battery: 77 | Samples: 3  
Action: collect_sample | Position: (0, 0) | Battery: 72 | Samples: 4  
Action: move_forward | Position: (1, 0) | Battery: 69 | Samples: 4  
Action: move_forward | Position: (2, 0) | Battery: 66 | Samples: 4  
Action: navigate_around | Position: (2, 0) | Battery: 62 | Samples: 4  
Action: move_forward | Position: (3, 0) | Battery: 59 | Samples: 4
```

```
>>>
```

```
===== RESTART: C:/Users/DELL/AppData/Local/Programs/Python/Python313/qn1.py =====  
Total solutions found: 92
```

First solution:

```
Q . . . . .  
. . . . . Q . . . .  
. . . . . . Q  
. . . . . Q . .  
. . Q . . . . .  
. . . . . Q .  
. Q . . . . .  
. . . Q . . . . .
```

```
>>>
```

```
===== RESTART: C:/Users/DELL/AppData/Local/Programs/Python/Python313/qn1.py =====  
Recommended cab: SHARED | Estimated Fare: Rs.85 | ETA: 10 min  
Recommended cab: PRIME | Estimated Fare: Rs.316 | ETA: 3 min
```

```
>>>
```

```
===== RESTART: C:/Users/DELL/AppData/Local/Programs/Python/Python313/qn1.py =====  
Least-cost path: S -> A -> C -> G  
Minimum cost: 4
```